

Classification Algorithm

Rules Set after training model:

Party A

score = **-11.1** pts $\leftarrow$ base points (starting score) + **0.039** $\times$ Age + **1.83** $\times$ Govt. satisfaction – **0.945** $\times$ Urban voter

Party T

score = **11.7** pts $\leftarrow$ base points (starting score) – **0.112** $\times$ Age – **1.71** $\times$ Govt. satisfaction + **1.21** $\times$ Urban voter

Party C

score = **0.985** pts $\leftarrow$ base points (starting score) + **0.115** $\times$ Age – **1.12** $\times$ Govt. satisfaction – **1.22** $\times$ Urban voter

Undecided

score = **1.50** pts $\leftarrow$ base points (starting score) – **0.009** $\times$ Age + **0.143** $\times$ Govt. satisfaction – **0.273** $\times$ Urban voter

a) Age <30 and Govt satisfaction less ,urban

MAKE A PREDICTION

Age 18.0 yrs

Govt. satisfaction 0 /10

Urban voter Yes

PREDICTED PREFERENCE **Party T**

	Step - 1									Step - 2a		Step - 2b	
	base point	Age		G_Satis		Urban		Value		e^Score where e = 2.718		% cal	
	Points	Points	Set Value	Points	Set Value	Points	Set Value	Manual	Model	Manual	Model	Manual	Model
Party_A	-11.1	0.039	18	1.83	0	0.945	1	-11.343	-11.3	0.000	0.000	0.00	0.00
Party_T	11.7	0.112	18	1.71	0	1.21	1	10.894	10.9	53852.279	54176.364	99.98	99.98
Party_C	0.985	0.115	18	1.12	0	1.22	1	1.835	1.83	6.265	6.234	0.01	0.01
Undecided	1.5	0.009	18	0.143	0	0.273	1	1.065	1.06	2.901	2.886	0.01	0.01

MAKE A PREDICTION

Age 30.0 yrs

Govt. satisfaction 2.50 /10

Urban voter Yes

PREDICTED PREFERENCE **Party T**

Party A
0.0%

Party T
97.4%

Party C
0.8%

Undecided
1.9%

	Step - 1									Step - 2a		Step - 2b	
	base point	Age		G_Satis		Urban		Value		e^Score where e = 2.718		% cal	
	Points	Points	Set Value	Points	Set Value	Points	Set Value	Manual	Model	Manual	Model	Manual	Model
Party_A	-11.1	0.039	30	1.83	2.5	0.945	1	-6.3	-6.26	0.002	0.002	0.00	0.00
Party_T	11.7	0.112	30	1.71	2.5	1.21	1	5.275	5.26	195.390	192.481	97.39	97.36
Party_C	0.985	0.115	30	1.12	2.5	1.22	1	0.415	0.415	1.514	1.514	0.75	0.77
Undecided	1.5	0.009	30	0.143	2.5	0.273	1	1.3145	1.31	3.723	3.706	1.86	1.87

b) Age >50 and Govt satisfaction less ,non urban

MAKE A PREDICTION

Age 82.0 yrs

Govt. satisfaction 0 /10

Urban voter No

PREDICTED PREFERENCE **Party C**

	Step - 1								Step - 2a		Step - 2b		
	base point	Age		G_Satis		Urban		Value		e^Score where e = 2.718		% cal	
	Points	Points	Set Value	Points	Set Value	Points	Set Value	Manual	Model	Manual	Model	Manual	Model
Party_A	-11.1	0.039	82	1.83	0	0.945	0	-7.902	-7.9	0.000	0.000	0.00	0.00
Party_T	11.7	0.112	82	1.71	0	1.21	0	2.516	2.47	12.379	11.822	0.04	0.04
Party_C	0.985	0.115	82	1.12	0	1.22	0	10.415	10.4	33356.235	32859.626	99.96	99.96
Undecided	1.5	0.009	82	0.143	0	0.273	0	0.762	0.734	2.143	2.083	0.01	0.01